

TracePulse – Architecture & Requirements Document

Section 1 – Overview

Standalone AI incident-ticket tool: RCA (root cause analysis) + similarity search. Ticket comes in – AI diagnoses it – system finds similar past resolved incidents – engineer reviews/fixes – resolution feeds back into future similarity matches. No dependency on other projects (Pulsegrid/LogPulse).

Section 2 – Stack

FastAPI, PostgreSQL + pgvector, Alembic, Groq API (openai/gpt-oss-120b) for RCA, sentence-transformers (all-MiniLM-L6-v2, CPU-only) for embeddings, Docker Compose (local), static API key auth. 100% free tier.

Section 3 – V1 Status: SHIPPED (tagged v1.0)

Phase	What It Does	Status	Key Issue Resolved
1. Scaffold	Docker Compose + FastAPI skeleton	DONE	health route cleanup
2. Tickets table + pgvector + Alembic		DONE	missing CREATE EXTENSION in migration
3. Plain CRUD + validation		DONE	clean pass
4. API key auth		DONE	weak AI-generated key replaced with real entropy (openssl)
5. RCA wiring (Groq)		DONE	spec'd model deprecated, swapped to gpt-oss-120b; fixed import
6. Similarity search (pgvector)		DONE	GPU/CUDA torch bloat fixed via correct --index-url pinning
7. Resolve endpoint		DONE	NULL-embedding crash in similarity query fixed
8. Seed script	18 realistic tickets across 6 domains, seeded DB	DONE	Real API calls not made causing 401s fixed
9. End-to-end ship gate		DONE	cold restart, all 4 endpoints verified, 29 tickets survived restart int

Section 4 – V1 Endpoints

POST /tickets, GET /tickets/{id}, GET /tickets, PATCH /tickets/{id}/resolve – all require X-API-Key header

Section 5 – V2 Roadmap (locked scope, phased)

Phase	Scope	Dependency	Status
2a	Incident triage (priority/severity/issue_type/team) + resolve endpoint	Needs dedicated machine	IN PROGRESS
2b	SLA management + monitoring/escalation	Needs background scheduler	Not started
2c	Engineer assignment + notification	Needs user table + notification channel	Not started
2d	Unified engineer dashboard (frontend)	Frontend stack TBD	Not started
2e	Email ingestion (IMAP/Graph)	Isolated, no dependency	Not started

Phase	Scope	Dependency	Status
Blocked	Pulsegrid webhook integration	Blocked on Pulsegrid's own deploy, spec ready	No ETA

Section 6 – Key Architectural Decisions

– Model swapped from spec'd Llama 3.1 (deprecated) to openai/gpt-oss-120b

– RCA/triage always fail-safe: null fields on timeout/error, ticket never fails to save

– 10s timeout, retry-once on bad JSON

– Alembic used from Phase 2 onward for all schema changes

– Seed data generated through real API calls so embeddings/RCA are genuine, not placeholder

Section 7 – Deployment

Local Docker Compose only (2 containers: api, db). No cloud deploy yet.